

Heat Exchanger Correlations for Hydrogen Aviation

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Abstract

Future hydrogen-powered jet engines require compact, low-pressure-drop heat exchangers with wide transverse tube spacings (e.g., 6 diameters between tube rows, $X_t^* = 6$). These spacings fall outside of valid geometric limits of the experimental correlations that exist in literature, raising the uncertainty of the use of the correlations. This study uses CFD to first try to match the empirical correlations and then, once validated, extends the transverse spacing to the specification that would be required for future heat exchangers. Following validation against published models by Gaddis and Gnielinski and data points provided by Kays and London, results demonstrate that extrapolating current correlations to wider spacings overpredicts both the Stanton number and friction factor by nearly an order of magnitude. The simulations highlight that modelling a minimum of three tube lanes is essential better to capture transient wake behaviours, which simplified half-lane models fail to resolve. Ultimately, this study suggests new, piecewise log-linear correlations tailored to $X_t^* = 6$ geometries and identifies an optimal operating point $Re_d \approx 1,250$ to maximise the heat transfer to pressure drop ratio, providing predictive tools for hydrogen aviation.

1 Introduction

Future hydrogen-powered jet engines will require compact and low pressure drop heat exchangers to facilitate new and advanced cycles. The transition to liquid hydrogen as a commercial aviation fuel offers a promising pathway to low-carbon air transport and provides substantial reductions in take-off weight due to hydrogen's high gravimetric heat of combustion. To maximize the thermal efficiency of these novel turbofan engines, designs typically integrate exhaust gas-to-hydrogen heat exchangers, located downstream of the low-pressure turbine, to heat the gaseous fuel prior to injection.

However, these specific aerospace applications necessitate tube bank designs with uniquely large transverse spacings, up to 6 diameters between the tube rows, which are outside the geometric boundaries of standard empirical correlations. Thus, numerous graphical and empirical correlations exist for land-based applications, such as those established by Kays and London or Gaddis and Gnielinski, it remains unknown to what extent these existing models are translatable to the extended geometries required for hydrogen aviation. Extrapolating these models beyond their intended limits introduces significant uncertainty in predicting the two main criteria of heat exchanger performance: the amount of heat transfer (characterized by the Stanton or Nusselt number) and the pressure drop (characterized by the friction factor).

To bridge this knowledge gap, this study investigates the flow behaviour and performance of tube banks at extended transverse spacings using 2D Unsteady Reynolds-averaged Navier-Stokes (URANS) CFD simulations. First the simulation methodology is validated against established correlations for conventional, tighter spacings, the computational models are then extended to replicate the $X_t^* = 6$ geometry proposed for hydrogen aircraft. Ultimately, this research evaluates the limitations of extrapolating existing literature and derives new, accurate predictive correlations specifically tailored to the design of fuel heat exchangers for future aerospace applications.

2 Background

Although there are many correlations in literature for land-based applications, they are still commonly applied to aerospace applications beyond their valid geometric ranges. However, it is not known to what extent that the existing correlations are translatable to the conditions of future aerospace applications,

in particular, this would involve increased transverse spacing between tube bank heat exchangers placed in cross-flow, beyond the existing transverse spacings used in correlations such as those produced by Kays and London [1] or by Gaddis and Gnielinski [2]. These large transverse spacing tube banks would be able to reduce the pressure drop across the heat exchanger, which is especially important for those placed within the exhaust flow of a jet engine, used to heat the hydrogen before combustion.

To evaluate the performance of a heat exchanger, there are two main metrics that reflect how well a heat exchanger is suited to an application: the amount of heat transfer and the pressure drop across the heat exchanger. The heat transfer is typically characterised using a dimensionless form such as the Nusselt number or Stanton number, with the former quantifying the ratio of convective heat transfer to conductive heat transfer, as in Eqn.1. The latter quantifies the ratio of heat transferred into a fluid to the thermal capacity of the fluid, as in Eqn.2. The Stanton number can also be related to the Nusselt number by the Reynolds number, Re , and the Prandtl number, Pr . The pressure drop can be characterised using the relative pressure drop across the heat exchanger $\Delta P/P_{in}$ but also by various dimensionless forms that quantify the amount of viscous effects, the friction factor f , being the most common.

$$Nu = \frac{hd}{\lambda} \quad (1)$$

$$St = \frac{h}{\rho U c_p} = \frac{Nu}{Re Pr} \quad (2)$$

The performance of heat exchangers can be found from experimental methods, and measuring the required quantities, or by using empirical or graphical correlations based on experimental results, that only require the geometry of the heat exchanger along with the flow parameters.

In order to produce a set of correlations that would be applicable to hydrogen aviation, it was desirable to first replicate the correlations produced by Kays and London using URANS CFD simulations to confirm that the models could be used reliably, and then change the geometry to match the spacings that would be used in the fuel heat exchanger of a hydrogen jet engine. This was done by reproducing the experiments done by Kays and London at the spacings and conditions described in 'Compact Heat Exchangers' and comparing the plotted results gathered in the CFD simulations to the graphical correlations. This method would then be extended to other correlations to further check the validity of the simulation model.

2.1 Literature Review

Kays and London [1] provide a set of graphical correlations for the pressure drop and heat transfer for both inline and staggered tube-banks placed in cross flow. These correlations are based on a set of experiments that utilised a wind tunnel in which test cores were placed. Each test core was 8.375 in by 9.75 in when viewed from the front, with the length of the test core set by the longitudinal space required to fit 15 tube rows, each of outer diameter 0.375 in (d_o). Plastic tube sheets were placed at the top and bottom of the test core to act as walls, to ensure the flow travelled within the test core. These test cores were used to heat up the incoming flow of air over the tubes, which were held at a constant temperature by passing superheated steam through the test core tubes in the steady state test conditions.

The correlations produced by Kays and London from the steady state tests exist for a transverse spacing (X_t) of $1.5 d_o$, and a longitudinal spacing (X_ℓ) of $1.25 d_o$ between the tubes, for both inline and staggered conditions, and give information for the pressure drop and heat transfer based solely on the geometry and Reynolds number of the external flow.

From the correlations, the pressure drop is characterised by the dimensionless number called the friction factor f , which combines the inlet and outlet effects of the tube bank along with the form drag introduced by the geometry of the heat exchanger, being calculated by Eqn.3.

$$\frac{\Delta P}{P_1} = \frac{G^2}{2g_c} \frac{v_1}{P_1} \left[(1 + \sigma^2) \left(\frac{v_2}{v_1} - 1 \right) + f \frac{A}{A_c} \frac{v_m}{v_1} \right] \quad (3)$$

The heat transfer of the heat exchanger is characterised by the dimensionless group of $St Pr^{\frac{2}{3}}$ where each dimensionless number is calculated using Eqn.4 and Eqn.5 respectively. The dimensionless group of $St Pr^{\frac{2}{3}}$ provides a metric for the performance of the heat exchanger, and allows for the Nusselt number to be calculated.

$$St = \frac{h}{G c_p} \quad (4)$$

$$\text{Pr} = \frac{\mu c_p}{\lambda} \quad (5)$$

These correlations are accurate to be within $\pm 5\%$ for the Prandtl number, friction factor, and $\text{St Pr}^{\frac{2}{3}}$, while the Stanton number alone is accurate within $\pm 3.5\%$.

Shah [3] provides empirical correlations directly for the pressure drop and Nusselt number for a wider range of geometric configurations compared to Kays and London, but only up to a transverse spacing $X_t = 3(d_o)$. These correlations are an expansion of those produced by Gaddis and Gnielinski [2] and later worked on by Martin [4] who generalised the L  v  que equation for heat exchangers. These correlations are based off of two equations in Eqn.6 and Eqn.7; which utilise two additional functions, the Hagen number Hg which is a function of the Reynolds number and geometry of the flow, and the L  v  que number Lq which is a dimensionless group used in fluid mechanics and heat or mass transfer to relate convective transfer rates to fluid friction and flow geometry. These correlations have a validity range of a transverse and longitudinal spacing up to $X_t^* = X_\ell^* = 3$.

$$\Delta P = \frac{\mu^2}{\rho g_c} \frac{N_r}{d_o^2} Hg \quad (6)$$

$$\text{Nu} = \begin{cases} 0.404 Lq^{1/3} \left(\frac{\text{Re}_d + 1}{\text{Re}_d + 1000} \right)^{0.1} & \text{inline tube bundles} \\ 0.404 Lq^{1/3} & \text{staggered tube bundles} \end{cases} \quad (7)$$

$$f = \frac{2Hg}{\text{Re}_d^2} \quad (8)$$

A lesser used correlation is provided by Gunter and Shaw [5]. This correlation provides the friction factor of a heat exchanger in the form of a graphical correlation that does not depend on the arrangement of the heat exchanger; achieved by applying dimensional analysis to the geometry and utilising the volumetric hydraulic diameter D_v . The correlation produces a 'master curve' that when plotted on log-log axes is two straight lines, one for the laminar regime and one for the turbulent regime, with equations for each line shown in Eqn.9, which give a modified half-friction factor $\frac{f'}{2}$. The half-friction factor can be converted to the Fanning friction factor f_F by multiplying by the Sieder-Tate viscosity ratio, Eqn.10, which equates to unity when the dynamic viscosity remains constant across the temperature difference between the wall and bulk fluid.

$$\frac{f'}{2} = \begin{cases} 90 \cdot \text{Re}_d^{-1} & \text{Re} < 200 \text{ (Laminar Regime)} \\ 0.96 \cdot \text{Re}^{-0.145} & \text{Re}_d > 200 \text{ (Turbulent Regime)} \end{cases} \quad (9)$$

$$\frac{f'}{2} = f_F \cdot \left(\frac{\mu}{\mu_{\text{wall}}} \right)^{0.14} \quad (10)$$

While experimental heat exchanger correlations are widely available in literature, there was a scarcity of papers that specifically looked at modelling tube bank heat exchangers using CFD simulations. The related paper found that used CFD to simulate tube bank heat exchangers was by Kot  m  d and Brodniansk   [6], although it looked at attempting to identify the reference temperature used to calculate the Nusselt number. Furthermore, there are very few papers that consider the transition region from laminar to turbulent flow in the correlations produced by Kays and London, leading to uncertainty about the shape of the correlations, in particular the presence of a 'dip' within the graphical correlations.

Kot  m  d and Brodniansk   used both DNS and $k - \omega$ SST solvers to identify the reference temperature used to calculate the Nusselt number that was used in many correlations, by comparing their computational results to the correlations. This is important as depending on the choice of the reference temperature can cause an offset in the value of the Nusselt number, in some cases by a large amount. For the application of this research, this paper provided a large number of correlations that could be used to verify the simulation results. In their computational experiment, DNS was used for the laminar flow regimes, while the $k - \omega$ solver was used in the turbulent regimes, modelling an inline and staggered tube banks with $X_t^* = X_\ell^* = 2$, defining the Reynolds number and Prandtl number as in Eqn.16b and Eqn.5 respectively for each of the correlations. The correlations used for comparison come from Grimson [7],   kauskas [8], Yilmaz [9], Hausen [10] and Khan [11] and for the above geometry are reproduced below

in Eqns.11 – 15 respectively. For the case of air flowing over isothermal tubes, at sub-sonic velocities, the value of $\frac{Pr}{Pr_w}$ equates to unity as the Prandtl number at the wall is the same as that of the mean flow.

$$\text{Grimison : } Nu = 1.13 \cdot 0.229 Re_d^{0.632} Pr^{1/3} \quad (11)$$

$$\text{Zukauskas : } Nu = \begin{cases} 0.52 Re_d^{0.5} Pr^{0.36} \left(\frac{Pr}{Pr_w} \right)^{1/4} & Re_d = 10^2 - 10^3 \\ 0.27 Re_d^{0.63} Pr^{0.36} \left(\frac{Pr}{Pr_w} \right)^{1/4} & Re_d = 10^3 - 10^4 \end{cases} \quad (12)$$

$$\text{Yilmaz : } Nu = 0.9 Re_d^{0.4} \left[1 + \left(\frac{Re_d}{290} \right)^5 \right]^{0.5} Pr^{0.36} \quad (13)$$

$$\text{Hausen : } Nu = \left(1 - 0.0605 \sqrt{\frac{1000}{Re_d}} \right) \cdot 0.34 Re_d^{0.61} Pr^{0.31} \quad (14)$$

$$\text{Khan : } Nu = 1.43 \left(0.542 Re_d^{1/2} Pr^{1/3} + 0.001 Re_d \right) \quad (15)$$

An issue that was encountered when comparing different correlations was the different definitions used to define the dimensionless numbers. One key example of this was the definition of the Reynolds number, with Kays and London using a Reynolds number based on the hydraulic diameter (where $D_h = 4 r_h$), which was used as this definition also applied to complex three-dimensional structures where the cross-section is changing along the length of the flow. In contrast Gaddis and Gnielinski and the majority of the other correlations use a Reynolds number that is based on the tube outer diameter d_o . As a result when comparing the correlations the dimensionless numbers had to be recast so that the definitions were consistent throughout, using a Reynolds number that is based on the tube outer diameter, as this definition is the most prevalent in literature. Furthermore, the Nusselt number was recast to use the Stanton number to describe the heat transfer of the heat exchanger; which was chosen as the Stanton number directly links the fluid friction to the heat transfer.

$$Re = \frac{G D_h}{\mu} \quad (16a)$$

$$Re_d = \frac{\rho U_m d_o}{\mu} \quad (16b)$$

Systematic evaluations of liquid hydrogen as a commercial aviation fuel was pioneered by Brewer, [12], demonstrating that despite requiring approximately 4.15 times the fuel volume, due to its low density and 2.8 time higher gravimetric heat of combustion, liquid hydrogen results in a substantial take-off weight reduction, along with the added benefit of providing a pathway to low carbon air-transport. Brewer's integration of an exhaust gas-to-hydrogen heat exchanger maximised the thermal efficiency of hydrogen fuelled turbofan engines. Placing an inline tubular heat exchanger downstream of the low-pressure turbine to heat the gaseous fuel before injection reduced the specific fuel consumption. This heat exchanger had a transverse spacing $X_t = 6 d_o$ which placed it outside of the geometric ranges of most of the heat exchanger correlations. The single operating point could be back calculated to yield values for the Reynolds number and friction factor, providing a data point for comparison, $Re = 3000$, $f = 0.0314$.

3 Simulation Setup

3.1 Computational Constraints

Computational power and limited time was a key factor to the choice to run all the simulations in 2D, and to use RANS simulations methods, as LES and DNS would require significantly greater computational requirements along with a greater amount of time. Initially all simulations were run on a machine with 6 cores and 10 GB of unified memory. This was fine for smaller mesh sizes and the small number of simulations at the start, however, this became a bottleneck in the progress of the research. As a result, the simulations were then ran on a desktop with a 32 core CPU and 128 GB of RAM, which led to simulations running on average three times faster. Moving over to the desktop also allowed the

simulations to run for longer, as batches of simulations could be ran over night, allowing for better time utilisation.

Comparing the run time between a 3D and 2D simulation of a staggered tube bank, in Table.1, it is clear that the 2D simulation - due to the reduction in mesh size by a factor of 6 - takes a quarter of the time for the same 5000 iterations. In addition, the results from the simulations are comparable, as the 3D behaviour of the flow did not affect the heat transfer or pressure drop significantly when the inlet was set to be a uniform velocity-inlet. This drastic reduction in run time is greatly important, considering the number of simulations required to plot each of the correlations, with 40 simulations being run for a sweep of Reynolds number for each tube bank geometry.

Table 1: Model comparison showing mesh size and run time.

Model	Number of cells	Run time
2D	296,469	45 mins
3D	1,997,389	3 hrs 4 mins

3.2 Domain Setup

To model the tube bank in CFD, the boundary conditions of the domain were set to replicate physical experiments that were done by Kays and London, however, using a different geometry of heat exchanger, as shown in Fig.1. As such the inlet was prescribed to have an inlet velocity of U_{in} which was varied to produce a sweep of Reynolds number, along with a static temperature $T_{in} = 300$ K to replicate the ambient air temperature. The outlet of the domain was prescribed as a pressure outlet, with the average pressure across the outlet plane set to $P_{atm} = 101,325$ Pa, where the average pressure condition allows the pressure to vary across the outlet plane as would be expected physically due to the formation of jets between the tube rows. For the top and bottom of the domain, symmetric boundary conditions were chosen, requiring fewer tube rows to be modelled, reducing the mesh size and run time. Finally, the tube walls were prescribed to be isothermal with a temperature of $T_{wall} = 373.15$ K replicating the steady-state tests conducted using superheated steam in order to maintain a constant temperature of the boiling point of water as the tube wall temperature. The geometric and boundary condition parameters of the domain are provided in Table.2.

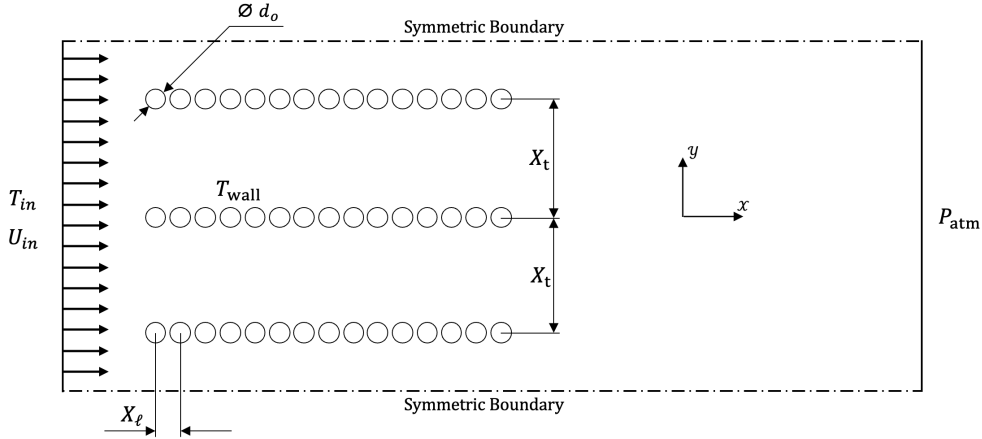


Figure 1: Diagram of the simulation domain.

Table 2: Domain parameter descriptions and dimensions.

Parameter	Description	Dimension
d_o	Tube Outer Diameter	0.00635 m
X_t	Transverse Spacing	$6 d_o$
X_ℓ	Longitudinal Spacing	$1.25 d_o$
T_{in}	Inlet Temperature	300 K
T_{wall}	Tube Wall Temperature	373.15 K
P_{atm}	Atmospheric Pressure	10, 1325 Pa
U_{in}	Inlet Velocity	Variable

3.3 Mesh Independence Study

A mesh independence study was performed to calculate the minimum mesh size required, in order to reduce the run time of simulations, while still providing results that are representative of physical flow behaviours. The base cell size was decreased from $d_o/2$, shown in Fig. 2a to $d_o/10$, shown in Fig.2c and flow parameters were evaluated and compared to the finest mesh produced, as shown in Table.3. It was desirable for the mesh to have a percentage difference of less than 1% when compared to the finest mesh, as such the base size of $d_o/5$ was chosen, shown in Fig.2b, providing a compromise between the total cell count of the mesh and being able to capture the flow behaviour accurately, Fig.3.

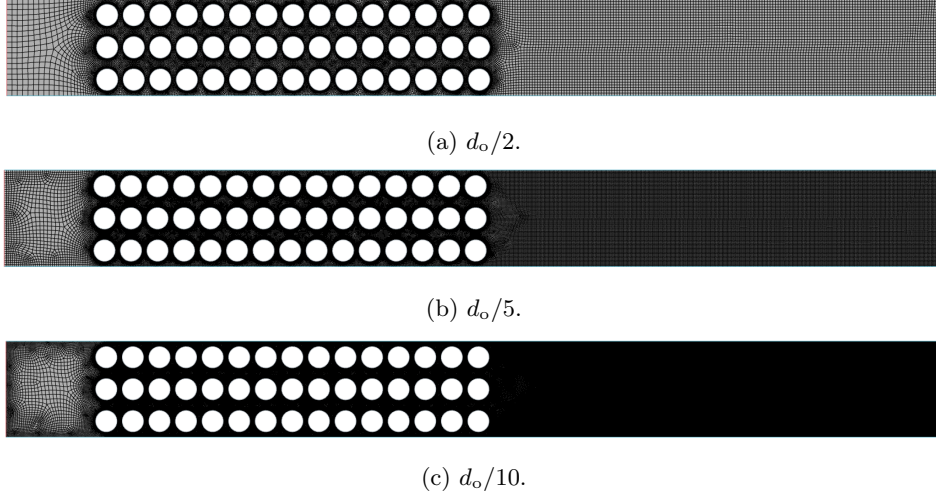


Figure 2: Three mesh resolutions used in the mesh independent study.

Comparing the run time between the 5 coarsest meshes, the relationship is not quite linear, however, there is a clear and significant increases in run time as the mesh becomes more refined. Run time values for finer meshes up to $d_o/10$ are omitted due to different computational hardware being used to run those simulations.

Table 3: Mesh independence study results.

Mesh Size	h	% Difference	Nu	% Difference	P_{in} (Pa)	% Difference	St	% Difference	T_{out} (K)	% Difference	Run time (s)
D/1	50.3564	47.58%	12.2842	47.58%	87.3479	58.45%	0.00630515	42.92%	339.134	3.98%	5736
D/2	36.4173	6.73%	8.88381	6.73%	58.7747	6.62%	0.00468755	6.26%	328.098	0.60%	10853
D/3	34.4762	1.04%	8.41027	1.04%	55.8269	1.27%	0.00443923	0.63%	326.419	0.08%	14675
D/4	34.3619	0.71%	8.38241	0.71%	55.3753	0.45%	0.00446346	1.18%	326.303	0.05%	17823
D/5	34.3944	0.80%	8.39033	0.80%	55.5395	0.75%	0.00444431	0.74%	326.451	0.09%	21500
D/6	34.2348	0.33%	8.35141	0.33%	55.4839	0.65%	0.00442420	0.29%	325.843	0.09%	
D/8	34.4556	0.98%	8.40526	0.98%	56.1576	1.87%	0.00445137	0.90%	326.224	0.02%	
D/10	34.1213		8.32372		55.1250		0.00441156		326.146		

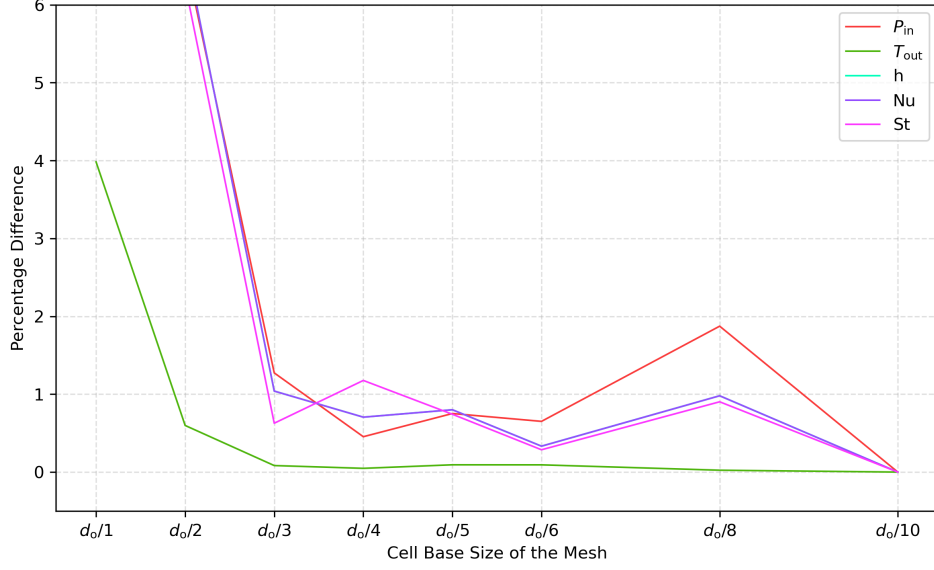


Figure 3: Percentage difference of flow parameters compared to finest mesh.

Looking closer at the prism layers around the tube walls, Fig.4, there is substantial mesh refinement around each of the tube walls, which allows for better modelling of the boundary layers formed around the surface of each tube.

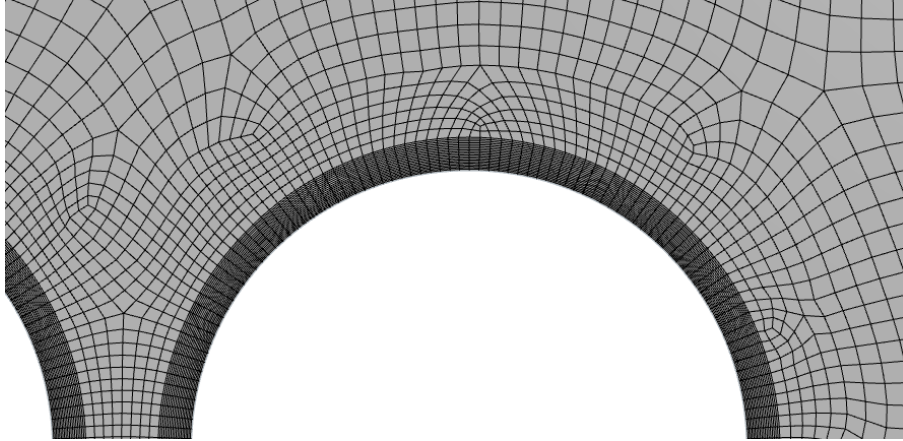


Figure 4: Prism layers of $d_o/5$ mesh around the tube walls.

3.4 Turbulence Models

Within Star-CCM+ there are various different RANS turbulence models, with two models being predominantly used for URANS simulations, those being $k - \epsilon$ and $k - \omega$ turbulence models. Both have their advantages and disadvantages. Additionally, laminar models are available to model the laminar flow regime at low Reynolds numbers.

At low Reynolds numbers the flow within the heat exchanger is expected to be laminar, however, when comparing the use of laminar and turbulent models at these low Reynolds numbers both performed similarly. At higher Reynolds numbers the results between the two models began to diverge, with the turbulent model producing results closer to the correlations. Consequently, to reduce the number of unique simulations that would be required, all of the simulations were ran using turbulent models, as the performance is comparable to the laminar models at low Reynolds numbers.

The $k - \epsilon$ turbulence model solves the governing transport equations for k (kinetic energy) and ϵ (the rate of dissipation of turbulent kinetic energy). The $k - \omega$ turbulence model similarly solves for k but instead solves for ω (the frequency of large eddies). The main advantage of using the $k - \omega$ turbulence model is its performance when resolving near-wall flows, especially in the presence of strong adverse

pressure gradients, whereas the $k - \epsilon$ model is often inadequate at resolving the flow at the near-wall regions and requires the use of wall-functions to bridge the viscous sublayer. In contrast the $k - \epsilon$ model is less sensitive to free-stream boundary conditions, making it more robust during initial problem setup.

However, the shear-stress transport (SST) $k - \omega$ turbulence model proposed by Menter, as outlined in [13], is a hybrid formulation that merges the strengths of both $k - \omega$ and $k - \epsilon$ turbulence models. For near-wall regions, the $k - \omega$ model is used to effectively handle any adverse pressure gradients and the boundary layers at the walls of the domain; while as distance from the walls increases, the model gradually blends into the $k - \epsilon$ model, eliminating the free-stream sensitivity problems that the $k - \omega$ model faces in the far field of the domain. As a result, all of the simulations utilised the SST $k - \omega$ turbulence model.

4 Validation - Matching Existing Correlations

Before computing correlations for the extended spacings, where $X_t^* = 6$, it was important to confirm the simulation methods used were accurate. This was done by comparing the simulation results to existing correlations that exist in literature for heat transfer and pressure drop. Five different geometric layouts were simulated and the results compared against the existing correlations, of those three were inline tube banks, and two were staggered tube banks with their dimensions presented in Table.4 along with the correlations that they were compared against.

Table 4: Tube bank configurations used for simulations.

Configuration	X_t^*	X_ℓ^*	Correlations
Inline	1.5	1.25	Kays and London
Staggered	1.5	1.25	Kays and London, Gaddis and Gnielinski
Inline	3	1.25	Gaddis and Gnielinski
Staggered	3	1.25	Gaddis and Gnielinski
Inline	2	2	Grimison, Zukauskas, Hausen, Khan

4.1 $X_t^* = 1.5$, $X_\ell^* = 1.25$

4.1.1 Inline Tube Bank

The first correlations that were simulated were by Kays and London for an inline configuration with $X_t^* = 1.5$ and $X_\ell^* = 1.25$. Simulations were run for different numbers of lanes being modelled. The CFD simulation was not able to match the correlations in the transition region for the friction factor f or St in Fig.5. This was due to the simulation method not being able to accurately handle the turbulent transition of the flow. However, looking at the trends at $Re < 100$ and $Re > 10,000$, with Reynolds number in this case defined in by the hydraulic diameter, as in Eqn.16a, there is some agreement between the CFD simulations and the correlations. Considering the accuracy of the simulations, the full coupon and half coupon simulations - 22 and 11 tube lanes respectively - produced results that are closest to the correlation however, there is still a discrepancy to the correlation. Additionally, some of the simulations for the full and half coupon did not converge, which can be seen in Fig.5 by the anomalous points for the Stanton number. As a result the 3 lane and half lane simulations provided results that were comparable to the other simulations, while reducing the run time of the simulations drastically.

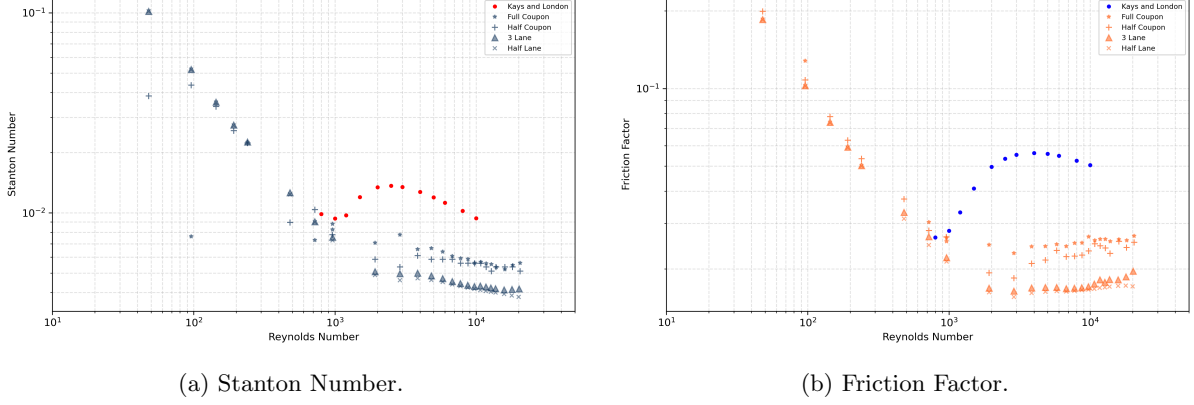


Figure 5: Comparison for inline $X_t^* = 1.5$, $X_\ell^* = 1.25$ against Kays and London.

4.1.2 Staggered Tube Bank

When simulating the same geometric spacing but using a staggered configuration, there was closer agreement to the correlations provided by Gaddis and Gnielinski along with Kays and London, with the friction factor being especially close to the Kays and London correlations, Fig.6b. Once again the simulation produces different results during where turbulent transition is expected to occur. An interesting observation between the two correlations plotted in Fig.6 is the fact there is a clear offset between them, with Kays and London predicting a higher Stanton number but lower friction factor.

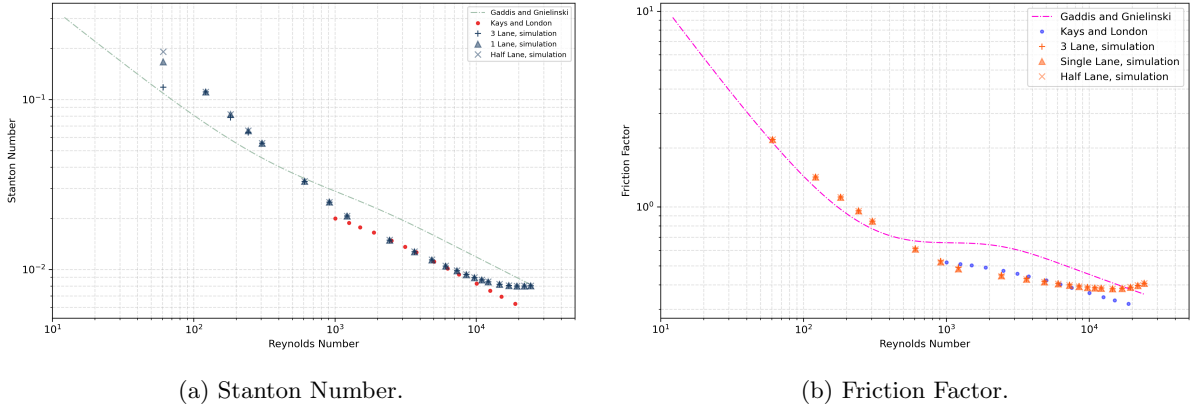
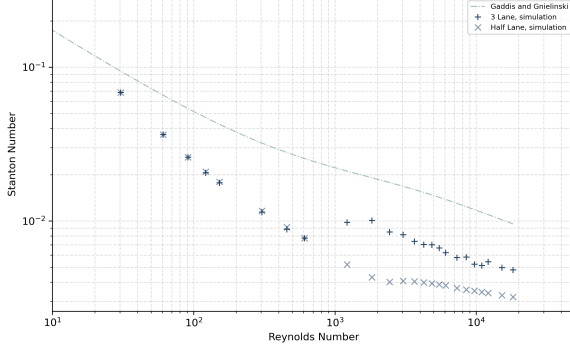


Figure 6: Comparison for staggered $X_t^* = 1.5$, $X_\ell^* = 1.25$ against Kays & London and Gaddis & Gnielinski.

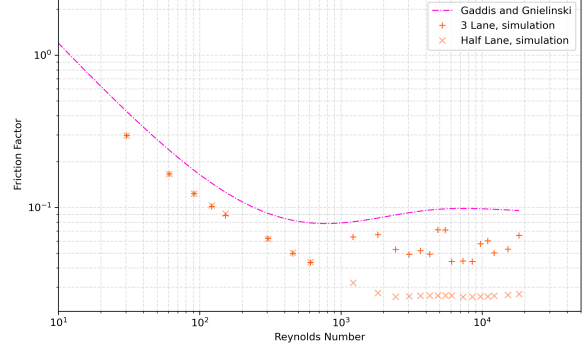
4.2 $X_t^* = 3$, $X_\ell^* = 1.25$

4.2.1 Inline Tube Bank

Extending the transverse spacing introduced some discrepancies between the 3 lane and half lane simulation for the inline tube bank. The larger transverse spacing caused the results of the two simulations to no longer agree for the entire range of Reynolds numbers, unlike for the simulations with $X_t^* = 1.5$. Compared to the correlations, there is a large offset, however for both plots, Fig.7, the 3 lane simulation underwent turbulent transition at a similar Reynolds number.



(a) Stanton Number.

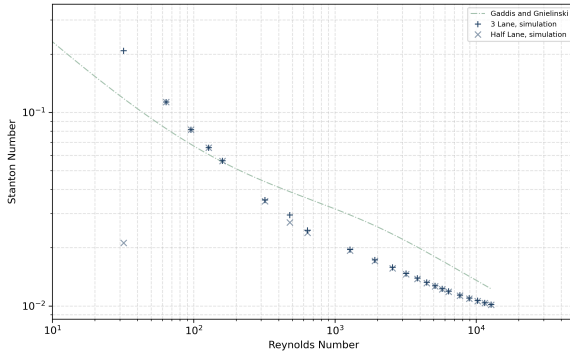


(b) Friction Factor.

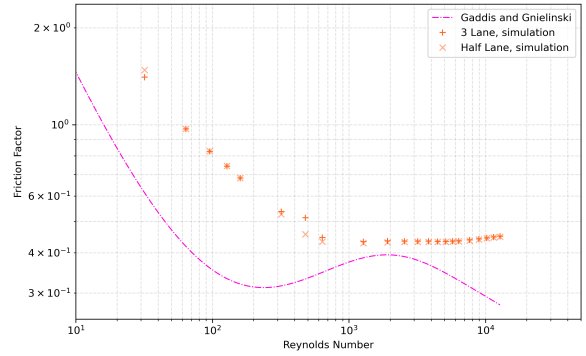
Figure 7: Comparison for inline $X_t^* = 3$, $X_\ell^* = 1.25$ against Gaddis & Gnielinski.

4.2.2 Staggered Tube Bank

Once again, when comparing the agreement to the correlations between the inline and staggered tube banks, for $X_t^* = 3$ the staggered tube bank simulation's results were closer to the Gaddis and Gnielinski correlations. However, compared to the inline simulations, the turbulent transition is not as well modelled, with no noticeable transition region present in Fig.8.



(a) Stanton Number.



(b) Friction Factor.

Figure 8: Comparison for staggered $X_t^* = 3$, $X_\ell^* = 1.25$ against Gaddis & Gnielinski.

4.3 Inline Tube Bank: $X_t^* = X_\ell^* = 2$

The final validation simulations that were run followed the methodology described in [6], modelling a half lane of the tube bank. From Fig.9, the simulation results are closest to the Žukauskas correlations, with close agreement at the two extremes of Reynolds number. Additionally, when comparing the various correlations against each other, there are different predictions for the Stanton number at lower Reynolds number, however at higher Reynolds number, almost all of the correlations converge, apart from the Žukauskas correlation and Gaddis and Gnielinski. Interestingly, the Gaddis and Gnielinski correlation over estimates the heat transfer by a large amount in comparison, but does show a change in the correlation around where turbulent transition is expected to occur.

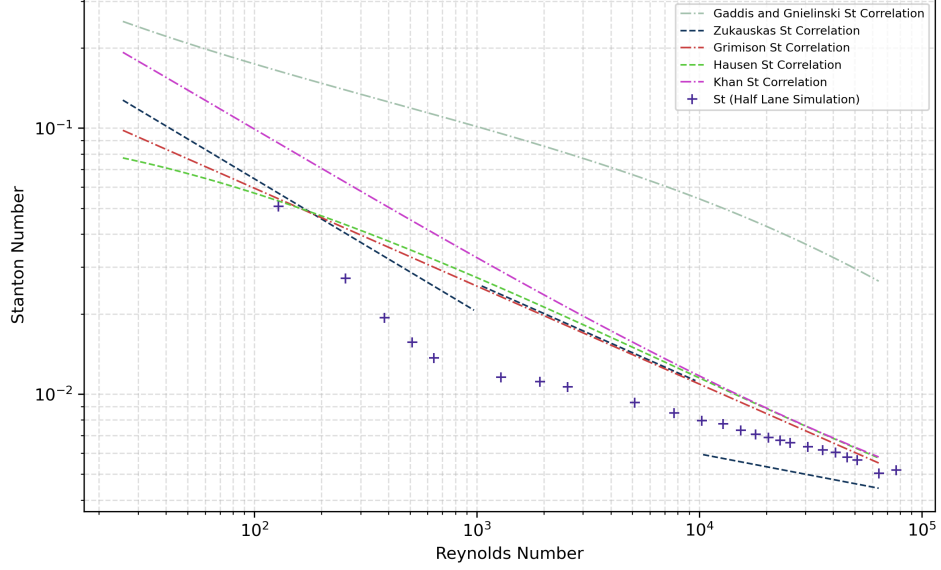


Figure 9: Comparison for inline $X_t^* = X_\ell^* = 2$ against various correlations.

5 Results & Discussion

5.1 Discussion from Validation Results

5.1.1 Inline vs Staggered

From the plots of the simulation results for the inline and staggered tube banks, it was clear that the results from the staggered simulations were closer to the predicted Stanton number and friction factor from the correlations compared to the inline simulations. One suggestion for this behaviour could be the unsteady behaviour of the jets that form between the lanes of tubes. In Fig.10, the jets that form between the tube rows exhibit unsteady behaviour once they leave the final tube row, with the jets coalescing into groups. The jets coalescing produced vortices in the wake of the tube bank.

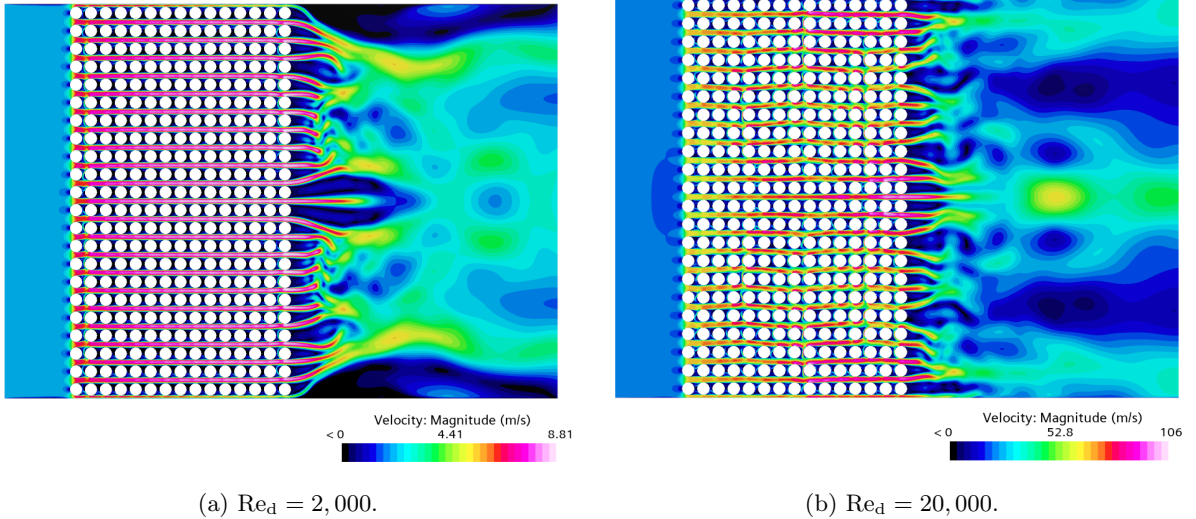


Figure 10: Velocity scenes for $X_t^* = 1.5$, $X_\ell^* = 1.25$ inline tube bank.

In contrast, the staggered tube banks in Fig.11 appear to not form vorticities downstream of the tube bank. At low Reynolds numbers, Fig.11a, the jets exit the last tube row straight and parallel before transitioning to a uniform velocity. At higher Reynolds numbers, Fig.11b, the jets begin to coalesce after

the last tube row, however, in comparison to the inline tube bank in Fig.10b, the merged jets appear to form a steady region in the wake of the staggered tube bank.

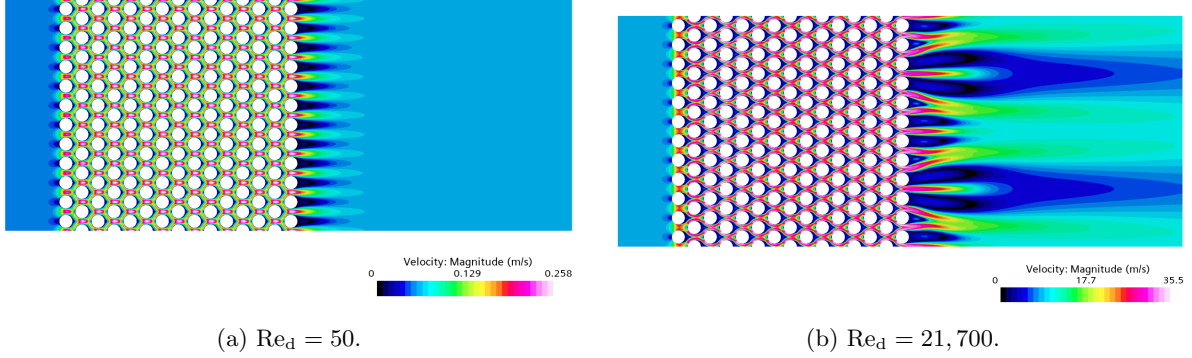


Figure 11: Velocity scenes for $X_t^* = 1.5$, $X_\ell^* = 1.25$ staggered tube bank.

As URANS simulations typically struggle with transient unsteadiness, it would suggest that the use of URANS simulation methods for the inline tube banks is not the best suited for modelling closer transverse spacings due to the vortex shedding that occurs as a result of the jets coalescing downstream. However, when comparing the simulation ran with $X_t^* = X_\ell^* = 2$, the larger spacing reduced the effect of the jets coalescing, and produced results that were closer to the correlations.

5.1.2 Number of Lanes

When comparing the simulations ran for the different number of lanes, initially for the $X_t^* = 1.5$, $X_\ell^* = 1.25$ simulations in Fig.5, there was close agreement between all of the simulations at lower Reynolds numbers, however, above the expected transition point at $Re_d = 2,000$, the full coupon and half coupon results begin to differ compared to the simulations of 3 lanes and half a lane. This suggests that there is a dependence on the Reynolds number how many lanes of a tube bank need to be modelled, and the boundary conditions placed onto the domain; as the domain for the full and half coupon simulations have walls on the top and bottom planes. As the results from this set of simulations were all far off from the correlation, simulations proceeded with only modelling 3 lanes or less.

When the transverse spacing was increased between the tube lanes, initially at low Reynolds numbers the 3 lane and half lane simulations once again agreed. Going above $Re_d = 1,000$ resulted in the results once again diverging, as in Fig.7, with the 3 lane simulations having a greater friction factor and Stanton number. However, this behaviour was not present in the staggered simulations with the same transverse spacing, Fig.8. Considering the steady wake produced by the staggered simulations, this result goes to suggest that there are transient behaviours of the flow that were not modelled in the half lane simulations, that were present in the 3 lane simulations, such as the jets coalescing. As a result, the half lane simulations that were modelled using the same methodology as in [6] should not be used in isolation.

5.2 Inline Tube Bank Extended Transverse Spacing $X_t^* = 6$, $X_\ell^* = 1.25$

Initially, only 3 lane and half lane simulations were ran, similar to the previous simulations ran during the validation. However, when comparing the results in Fig.12 and Fig.13 there was a more significant difference between the 3 lane and half lane simulations at higher Reynolds numbers compared to the validation simulations, which is especially clear in the friction factor plot. Consequently, further simulations were ran, modelling 2 lanes and a single lane to provide further data points, which confirmed that the half lane simulation results at higher Reynolds numbers were not modelling some of the flow behaviours that are present in a tube bank heat exchanger.

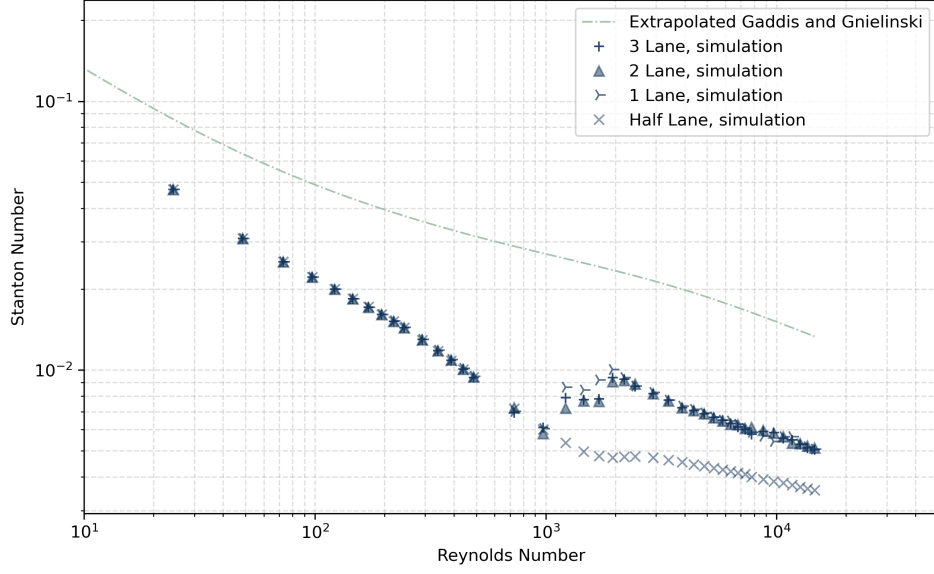


Figure 12: Inline $X_t^* = 6$, $X_\ell^* = 1.25$ comparison of Stanton number against Reynolds number.

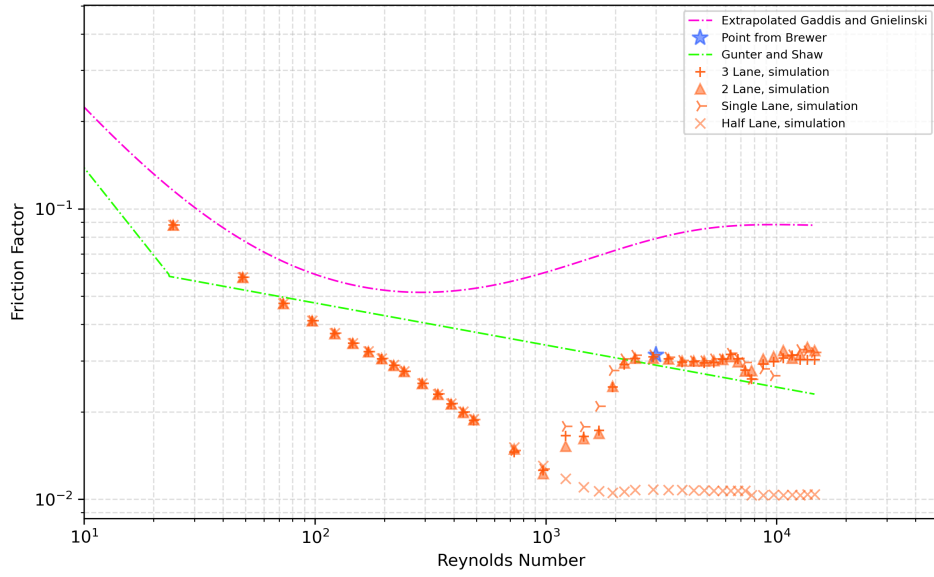


Figure 13: Inline $X_t^* = 6$, $X_\ell^* = 1.25$ comparison of friction factor against Reynolds number.

Looking at the temperature scenes for the tube bank, Fig.14, the flow becomes laminar unsteady for at $Re_d = 220$, Fig.14b and is transitional at $Re_d \approx 1,000$ as suggested by Fig.13. Fig.14c shows the flow within this turbulent transition region. At $Re_d = 2,000$ the flow around the tube bank is fully turbulent, with vortices beginning to form very early in the tube bank, as illustrated in Fig.14d which shows the formation of different scales of eddies at the highest Reynolds number simulated.

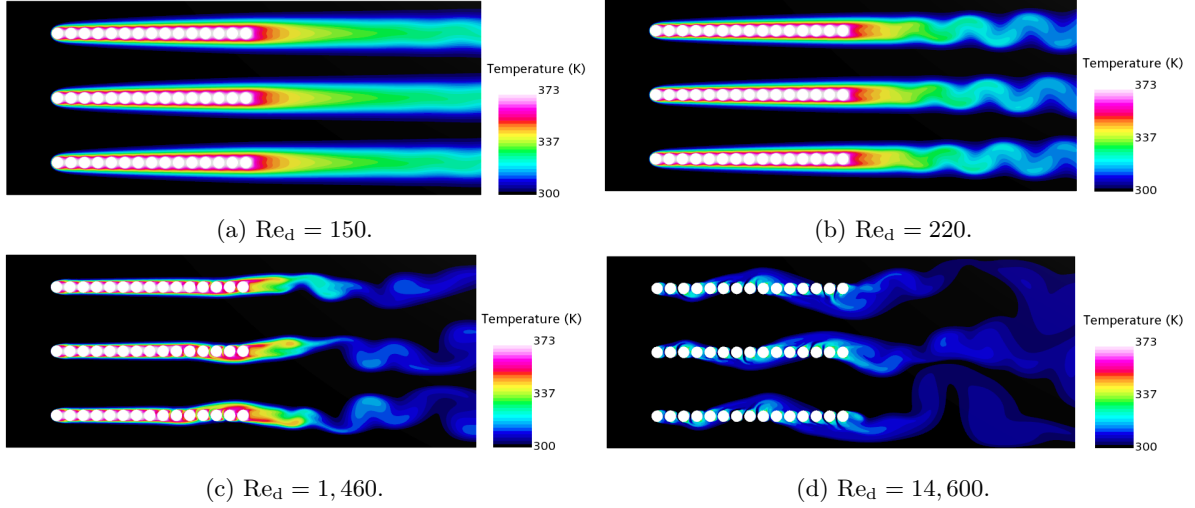


Figure 14: Inline $X_t^* = 6$, $X_\ell^* = 1.25$ temperature scenes for different Reynolds number.

The simulation results were able to replicate the single data point for the friction factor from Brewer, with the results from all apart from the half lane simulation matching Brewer’s data point, Fig.13.

5.2.1 Existing Correlations

The Gaddis and Gnielinski correlation is the only one found in literature with an increase in friction factor in the transition region, as seen in Kays and London’s experimental data of inline banks. The simulation results confirms that this increase occurs for the transverse spacings of $X_t^* = 6$ used in the hydrogen aircraft heat exchanger design. For Re_d between 200 and 1000, the friction factor is in fact 0.01 to 0.02 lower (25 – 50%) than predicted by the generic Gunter and Shaw correlation. The extrapolated Gaddis and Gnielinski correlations are included for completeness, but these should only be used for transverse spacings up to $X_t^* = 3$. From Fig.12 and Fig.13, there is a clear offset between the simulation results and the extrapolated correlations, predicting almost an order of magnitude greater Stanton number and friction factor.

5.2.2 Comparison to Gunter and Shaw

When comparing the simulation results against the correlation for the friction factor from Gunter and Shaw, it is clear that at Reynolds numbers below 1000 the correlation predicts transition much earlier, but at higher Reynolds numbers, the correlations proves to be very close at predicting the friction factor of the tube bank. The correlation is incidentally quite accurate at the Reynolds number of the Brewer design point.

The simulation results suggest that the use of this lesser utilised correlation is better suited for the application to heat exchangers with greater transverse spacing, such as those that would potentially be used in hydrogen jet engines, rather than extrapolating correlations that more frequently used. This was a surprising result due to the lack of reference to the Gunter and Shaw correlation in other literature, but could be coincidental.

5.2.3 Simulation Derived Correlations

From the simulation data, it was desirable to produce a set of correlations that could be used for calculating the Stanton number and friction factor across different Reynolds numbers. To achieve this, a log-linear model was used and fit to the data, as a piecewise function for above and below the turbulent transition region. The derived Stanton number correlation, Eqn.17 was chosen to be a function of Reynolds and Prandtl number, similar to how the traditional correlations predict the Nusselt number. The derived friction factor correlation, Eqn.18, is only a function of Reynolds number. After computing the derived correlations, they were plotted alongside the simulation data, with the Stanton number correlation plotted in Fig.15, and the friction factor correlation plotted in Fig.16.

$$\text{St}(\text{Re}_d, \text{Pr}) = \begin{cases} 0.298 \text{Re}_d^{-0.540} \text{Pr}^{1/3}, & 20 \leq \text{Re}_d \leq 1,000 \\ 0.0980 \text{Re}_d^{-0.299} \text{Pr}^{1/3}, & 2,000 \leq \text{Re}_d \leq 10,000 \end{cases} \quad (17)$$

$$f(\text{Re}_d) = \begin{cases} 0.435 \text{Re}_d^{-0.509}, & 20 \leq \text{Re}_d \leq 1,000 \\ 0.0297, & 2,000 \leq \text{Re}_d \leq 10,000 \end{cases} \quad (18)$$

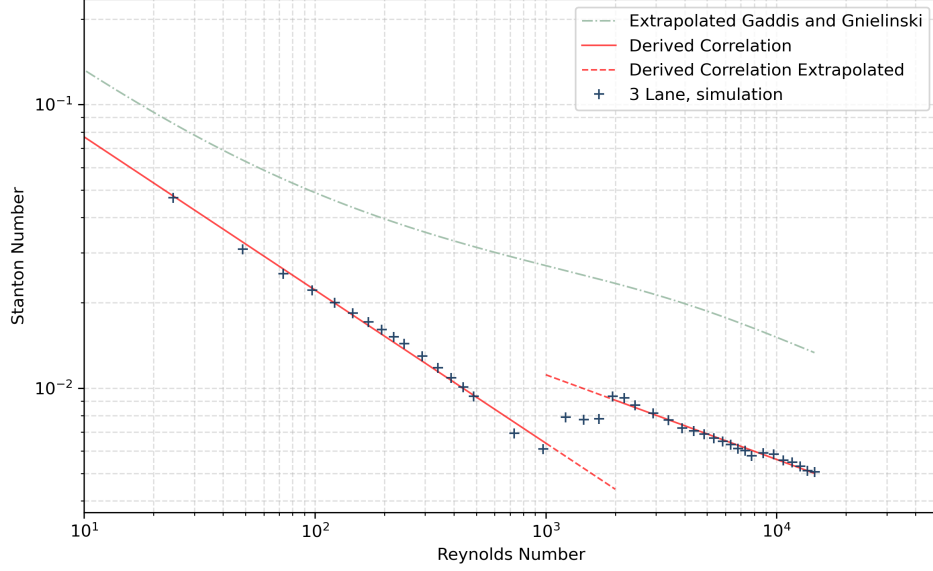


Figure 15: Derived Stanton number correlation plot.

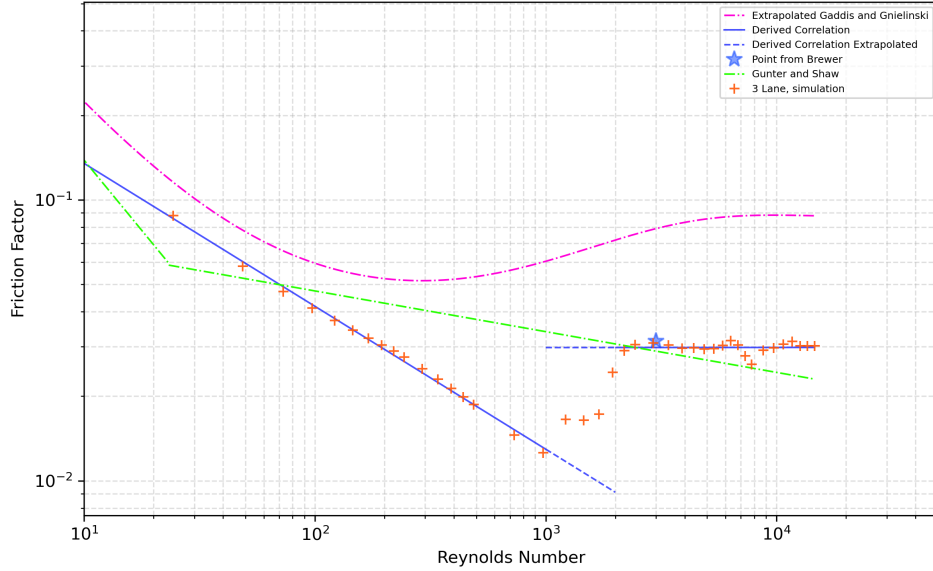


Figure 16: Derived friction factor correlation plot.

The derived correlations fit the simulation data very closely, with a coefficient of determination for the Stanton number and friction factor correlations being $R^2 = 0.997$ and $R^2 = 0.988$ respectively. Additionally, the derived friction factor correlation goes through the data point from Brewer. The transition region in both of the plots are plotted as dotted lines, extrapolating both parts of the piecewise function, as it is clear that the URANS simulation method struggles with turbulent transition.

5.2.4 Ratio of Heat Transfer to Pressure Drop

To quantify the performance of a heat exchanger, where a low pressure drop is advantageous, the ratio of heat transfer to the pressure drop across the heat exchanger can be used, where by maximising this ratio is optimal. This ratio, as defined in 19, can be plotted against Reynolds number to produce a curve, where the maximum suggests the operating point that maximises heat transfer while minimising the pressure drop.

Before computing j/f for the extended transverse spacing, to provide a reference to show how the plot is expected to look like, Fig.17 plots j/f for the staggered tube bank with $X_t^* = 1.5, X_\ell^* = 1.25$. It is expected that for a given Reynolds number, there exists a maximum, in this case at $Re_d = 1000$ where the ratio of heat transfer to the pressure drop is maximised.

$$j/f = \frac{St Pr^{2/3}}{f} \quad (19)$$

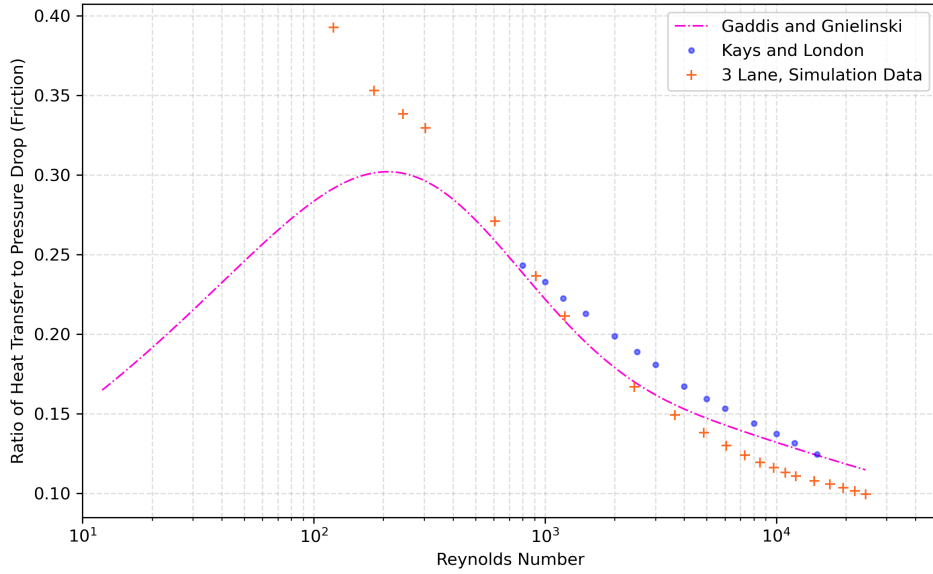


Figure 17: Ratio of heat transfer to pressure drop (friction) plot for staggered $X_t^* = 1.5, X_\ell^* = 1.25$.

Comparing Fig.17 to the plot of j/f for the inline tube bank with spacing $X_t^* = 6, X_\ell^* = 1.25$ in Fig.18, based on the derived correlation, the ratio could be increased drastically from the Brewer point by decreasing the Reynolds number from 3,000 to 1,000. When comparing the extrapolated Gaddis and Gnielinski correlation to the derived correlation, at low Reynolds numbers the heat transfer is predicted to be much greater, however, rather surprisingly, the two correlations are almost in agreement at higher Reynolds numbers.

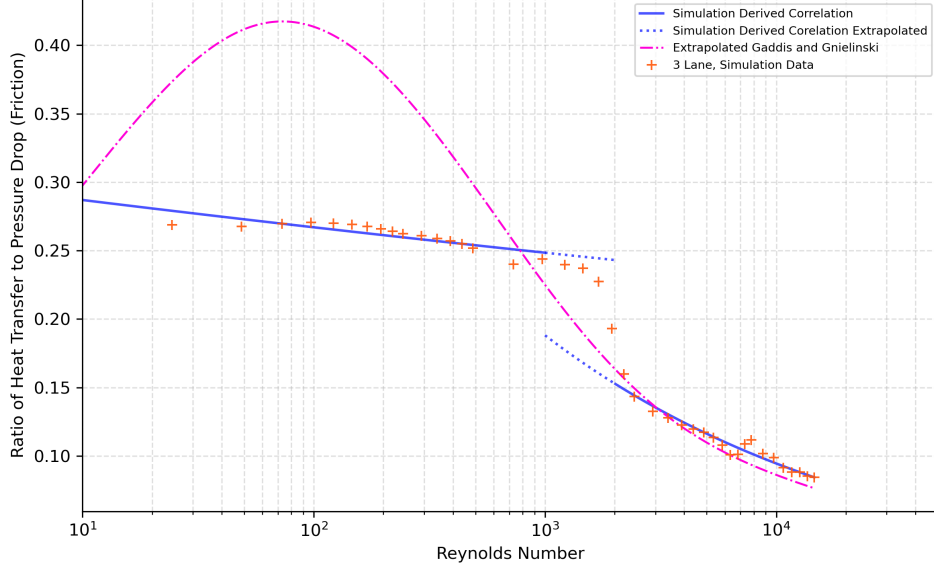


Figure 18: Ratio of heat transfer to pressure drop (friction) plot for inline $X_t^* = 6$, $X_\ell^* = 1.25$.

6 Conclusion

This study investigated the applicability of existing heat exchanger correlations to the extended transverse spacing required for future hydrogen-powered jet engines. Utilising 2D URANS CFD simulations, the flow behaviour and performance of tube banks were evaluated to determine how effectively standard correlations translate to a transverse spacing of $X_t^* = 6$. The findings drawn from the simulations highlight several critical insights into heat exchanger modelling and performance:

1. While URANS modelling struggles to accurately capture the laminar-to-turbulent transition region, it is otherwise effective; however, modelling at least three tube lanes is essential to properly capture transient wake behaviours and coalescing jets, as half-lane simulations proved inadequate at larger spacings and higher Reynolds numbers.
2. Extrapolating standard correlations, such as those by Gaddis and Gnielinski, beyond their geometric limits to $X_t^* = 6$ results in severe inaccuracies, over predicting both the Stanton number and friction factor by nearly an order of magnitude.
3. The lesser-utilised Gunter and Shaw correlation demonstrated surprising accuracy in predicting the friction factor at higher Reynolds numbers for extended spacings.
4. The multi-lane simulations successfully replicated the historical single data point from Brewer's hydrogen aircraft research.
5. New piecewise log-linear correlations, Eqn.17 and Eqn.18, for both the Stanton number and friction factor were successfully derived from the simulation data. These correlations offer predictive models tailored specifically for inline configurations with a transverse spacing of $X_t^* = 6$.
6. Evaluating the ratio of heat transfer to pressure drop (j/f) for the extended spacing revealed a local performance maximum near a Reynolds number of 1,250, indicating an optimal operating point for maximizing thermal transfer while minimizing pressure losses.

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9 Nomenclature

Table 5: Roman Letter Symbols.

A	Exchanger total heat transfer area
A_c	Exchanger minimum free-flow area
A_{fr}	Exchanger frontal area
c_p	Specific heat at constant pressure
D_h	Hydraulic diameter ($D_h = 4r_h$)
d	Characteristic tube dimension
d_o	Tube outer diameter
f	Friction factor
f_F	Fanning friction factor
G	Mass velocity
g_c	Proportionality factor in Newton's second law
h	Heat transfer coefficient
j	Non-dimensional heat transfer grouping ($St Pr^{2/3}$)
L	Heat exchanger flow length
N_r	Number of tube rows [stream-wise direction]
N_t	Number of tube lanes [transverse direction]
P	Static pressure
P_{atm}	Atmospheric pressure
P_{in}	Inlet pressure
R^2	Coefficient of determination
r_h	Hydraulic radius ($\frac{A_c L}{A}$)
T	Temperature
T_{in}	Inlet temperature
T_{wall}	Wall temperature
U	Velocity
U_{in}	Inlet velocity
v	Specific volume
X_t	Transverse spacing between tube lanes
X_t^*	Non-dimensional Transverse spacing between tube lanes ($\frac{X_t}{d_o}$)
X_ℓ	Longitudinal spacing between tube lanes
X_ℓ^*	Non-dimensional Longitudinal spacing between tube lanes ($\frac{X_\ell}{d_o}$)

Table 6: Greek Letter Symbols.

Δ	Difference
λ	Thermal conductivity
σ	Ratio of free-flow area to frontal area ($\frac{A_c}{A_{fr}}$)
μ	Dynamic viscosity
ρ	Density

Table 7: Dimensionless Groups.

Hg	Hagen number
Lq	L��v��que number
Nu	Nusselt number ($\frac{hd}{\lambda}$)
Re	Reynolds number ($\frac{GD_h}{\mu}$)
Re _d	Reynolds number ($\frac{\rho U_m d_o}{\mu}$)
St	Stanton number ($\frac{h}{\rho U c_p}$)
Pr	Prandtl number ($\frac{\mu c_p}{\lambda}$)

10 Appendix

Table 8: 3 Lane simulation data for $X_t^* = 6, X_\ell^* = 1.25$.

U_{in}	Re	St	f	U_{in}	Re	St	f
0.05	24.32	0.02410	0.0879	4.50	2188.83	0.00475	0.0291
0.10	48.64	0.01590	0.0580	5.00	2432.03	0.00447	0.0305
0.15	72.96	0.01300	0.0472	6.00	2918.44	0.00420	0.0310
0.20	97.28	0.01140	0.0412	7.00	3404.84	0.00397	0.0304
0.25	121.60	0.01030	0.0373	8.00	3891.25	0.00372	0.0297
0.30	145.92	0.00946	0.0344	9.00	4377.66	0.00363	0.0297
0.35	170.24	0.00881	0.0322	10.00	4864.06	0.00353	0.0295
0.40	194.56	0.00827	0.0305	11.00	5350.47	0.00343	0.0296
0.45	218.88	0.00781	0.0289	12.00	5836.87	0.00334	0.0303
0.50	243.20	0.00738	0.0275	13.00	6323.28	0.00325	0.0315
0.60	291.84	0.00667	0.0250	14.00	6809.69	0.00315	0.0304
0.70	340.48	0.00608	0.0230	15.00	7296.09	0.00309	0.0278
0.80	389.12	0.00559	0.0213	16.00	7782.50	0.00297	0.0260
0.90	437.77	0.00519	0.0199	18.00	8755.31	0.00303	0.0292
1.00	486.41	0.00481	0.0187	20.00	9728.12	0.00300	0.0297
1.50	729.61	0.00357	0.0145	22.00	10700.94	0.00286	0.0306
2.00	972.81	0.00314	0.0126	24.00	11673.75	0.00282	0.0313
2.50	1216.02	0.00406	0.0166	26.00	12646.56	0.00272	0.0302
3.00	1459.22	0.00398	0.0164	28.00	13619.37	0.00263	0.0303
3.50	1702.42	0.00401	0.0173	30.00	14592.18	0.00260	0.0302
4.00	1945.62	0.00481	0.0244				